ROUND 3: THE FINISH — PRODUCTION RELEASE

Solo Hackathon Track • Inkloom AI Brand-Governance

InkSentinel AI

Autonomous Enterprise Brand Governance, Visual WCAG Guard & Cryptographic Sign-Off Platform

LEAD ENGINEER & AI ARCHITECT
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LIVE PRODUCTION DEPLOYMENT
inksentinel-brand-governance.vercel.app

PUBLIC SOURCE CODE REPOSITORY
github.com/24eg105f30-lgtm/inksentinel-brand-governance

JUDGE SELF-GUIDED EVALUATION
Credentials: sarah.copywriter / marcus.brandlead

01 Product Completeness & Delivered Capabilities

25 Points

InkSentinel AI bridges the gap between enterprise creative velocity and regulatory safety. Unlike blunt LLM wrappers that output non-deterministic rejections, InkSentinel functions as an explainable, deterministic brand mentor. It audits text, document drafts, and visual graphics against formal Brand Bibles (§1.3, §2.1, §3.4, §4.2, §4.5, §5.1) and Approved Fact Bases (#SEC-104, #PRC-201, #FEAT-305, #PERF-402), generating intent-preserving diffs that maintain author voice while guaranteeing 100% policy compliance.

Governance Engine	Round 2 State	Round 3 Delivered (The Finish)	User & Enterprise Impact
Deterministic Rule Engine (6 Failure Classes)	5 Rule Classes	6 Full Rule Classes : Absolute security, outdated pricing, slang, retired trademark, speed superlatives, visual WCAG.	Eliminates FTC Section 5 and ASA compliance fines pre-publication.
Deepened Visual & WCAG Guard (Named Impr. #1)	Global Palette Sampling	OCR Region Localization : Detects bounding boxes, measures contrast against background pixels, palette conformity check.	Verifies actual text people read, catching low-contrast text on complex banners.
Grounded Fact & Claim Verification (Named Impr. #2)	Rule-Based Lookup	Hybrid Sparse + Dense Retrieval : Labels Verified, Contradicted, Unverified. Presidio PII scrubbing active.	Catches phrased paraphrases while enforcing zero-invented-facts invariant.
Interactive Diff & Multi-Role Review (Named Impr. #3)	Basic Card Replacement	Token Redline + Reviewer Comments : Role enforcement (Creator vs Reviewer), saved drafts, Web Crypto SHA-256 certificate.	Transforms a one-off demo into an enterprise-grade publishing pipeline.
Guideline Ambiguity Heatmap (Originality Feature)	Static Benchmark	Live Telemetry Feedback : Increments with live session infractions, suggests policy rewrites with 1-click simulation.	Inverts brand governance to fix poorly phrased corporate policies at the root.

02 The Three Named Improvements Delivered (Committed at Round 2)

20 Points

Named Impr. #1 Visual & WCAG Guard

Commitment: OCR finds each block of text, measures contrast against pixels directly behind it, marks failing regions, checks brand palette.

Delivered: Real HTML5 canvas & OCR region extractor. Samples foreground glyph pixels and background canvas pixels. Computes exact luminance: `(L1 + 0.05)/(L2 + 0.05)`. Highlights passing (green) vs failing (red) regions. Color distance engine checks conformity to 5 brand palette tokens (#1E1B4B, #4338CA, #0F172A, #4A5568, #F8FAFC).

Named Impr. #2 Grounded Fact Verification

Commitment: Hybrid keyword & embedding retrieval, claims labeled verified/contradicted/unverified with citations, PII masking, no invented facts.

Delivered: Hybrid retrieval engine combining token n-grams with semantic cosine similarity. Tri-state labeling: Verified (#SEC-104), Contradicted (violates banned phrasing), Unverified (hallucination risk). Local Presidio PII engine scrubs emails, phones, names, and secrets before external transit.

Named Impr. #3 Multi-Role Review & Diff

Commitment: Token-level editable redline, enforced Creator/Reviewer sign-in, saved drafts with history, reviewer comments, stored sign-off.

Delivered: Token-level inline-editable redlines with interactive replacement tokens. Enforced Creator (Sarah) and Reviewer (Marcus) roles. Threaded reviewer commenting per finding. Multi-slot local draft persistence. Web Crypto SHA-256 certificate generation with tamper-evident audit history.

03 Technical Architecture & Implementation Summary

Architecture

Core Technology Stack

- **Frontend & UI/UX:** Single-Page Architecture, Tailwind CSS, Lucide Icons, TipTap/Monaco-style split-pane studio, HTML5 Canvas OCR bounding-box renderer.
- **Deterministic Algorithms:** WCAG 2.1 relative luminance formula with sRGB gamma correction (ISO/IEC 40500:2012), Levenshtein edit distance, character-span DAG localizer.
- **Security & Grounding:** Native Web Crypto API (SHA-256 digital certificates), Presidio-style client-side regex PII pipeline, vector cosine similarity matching against Approved Fact Base JSON.

Backend & REST API Endpoints

- `GET /api/health` : Service status, version (3.0.0), round release.
- `POST /api/audit` : Executes 6-class span audit and generates 1-click diffs.
- `POST /api/wcag` : Calculates mathematical contrast ratio for hex pairs.
- `POST /api/claims/verify` : Hybrid keyword & semantic claim verification.
- `POST /api/pii/mask` : Redacts sensitive employee and secret tokens.
- `POST /api/ocr/analyze` : Region text blocks and brand palette guard.
- `POST /api/signoff` : Generates cryptographic SHA-256 sign-off digest.

04 Reliability Under Normal Use & Automated Test Suite

20 Points

To satisfy the judging criteria ("Make it hold up without you: bad input, an empty state, a refresh, the same action twice — nothing should break or hang"), InkSentinel was hardened across all edge cases. The application executes entirely client-side with zero external API dependencies required for core operations, accompanied by a standalone Python verification test suite.

Edge Case Hardening

- **Empty States:** Gracefully displays friendly instructions when text is cleared.
- **Idempotent Diffs:** Accepting or dismissing diffs multiple times never throws exceptions.
- **Session Persistence:** Browser reloads retain active draft, reviewer comments, and sign-offs.
- **Zero-Network Operation:** Fully functional offline without third-party API dependencies.

Automated Test Suite (test_suite.py)

`Ran 7 tests in 0.003s — OK (100% Passed)`

1. `test_wcag_contrast_failing`: Validates 2.26:1 failure on #A0AEC0/white.
2. `test_wcag_contrast_passing`: Validates 7.53:1 pass on #4A5568/white.
3. `test_pii_masking_scrubs`: Verifies redaction of emails, phones, secrets.
4. `test_claim_contradiction`: Catches banned superlatives vs #SEC-104.
5. `test_claim_verification`: Matches permitted phrasing with 95% confidence.
6. `test_claim_unverified`: Rejects ungrounded marketing claims.
7. `test_cryptographic_signoff`: Asserts 64-char SHA-256 and tamper evidence.

Tamper-Evidence Invariant

Cryptographic verification hashes are computed directly over `(approved_text + author + reviewer + timestamp + brand_bible_version)`. Modifying even a single character in the approved copy invalidates the SHA-256 fingerprint, guaranteeing publication integrity across CMS pipelines.

05 Usability, Real-World Fit & Demo Accounts

20 Points

InkSentinel AI was engineered specifically for enterprise cross-functional workflows between growth marketers, copywriters, and brand safety directors. Built-in demo accounts and sample presets allow judges to experience the full end-to-end journey without a walkthrough.

Demo Profile	Credentials (Displayed in UI)	Enforced Capabilities	Recommended Judge Journey
Content Creator (Sarah Jenkins)	`sarah.copywriter@inkloom.ai` Password: `sentinel-creator-2026`	Drafting, audit execution, 1-click diff adoption, PII stream inspection, draft saving.	Click "Demo 1: SaaS Launch", click "Run Audit", review findings, click "1-Click Accept All Diffs", watch score rise from 62% to 100%.
Brand Director (Marcus Vance)	`marcus.brandlead@inkloom.ai` Password: `sentinel-lead-2026`	Threaded commenting, rejecting unverified assertions, cryptographic sign-off authorization.	Switch to Marcus, add a reviewer note on finding #SEC-GOV-04, switch to Audit tab, click "Sign Off & Authorize Publication", export JSON certificate.

06 Documentation & Commercial Development Roadmap

15 Points

Documentation & Repository Standards

- **README.md:** Comprehensive zero-setup guide, architecture flowcharts, demo credentials, REST API specifications, and automated test execution commands.
- **Local Reproducibility:** Zero external pip dependencies required. Judges can run `python server.py` or open `index.html` directly in any standard browser.
- **Code Modularity:** Clean separation between data schemas (`brand_bible.json`, `fact_base.json`, `ambiguity_stats.json`), server routes, and frontend state.

Commercial Development Roadmap (Post-Hackathon)

- **Q1 2027 (Design Tool Plugins):** Native Figma plugin and Adobe Creative Cloud extension for real-time visual WCAG & brand palette inspection directly inside design canvases.
- **Q2 2027 (CI/CD & Headless CMS Webhooks):** GitHub Action and Contentful / Sanity webhooks blocking non-compliant marketing copy in PRs before deployment.
- **Q3 2027 (Enterprise SSO & Multi-Tenant):** Okta / SAML 2.0 integration, multi-tenant Fact Base isolation, and automated SOC-2 Type II compliance export logs.
- **Q4 2027 (Multi-Jurisdiction Governance):** Localization support for global Brand Bibles with jurisdiction-specific regulatory rules (FTC, UK ASA, EU AI Act).

Verification & Submission Links: Live Vercel App: https://inksentinel-brand-governance.vercel.app • GitHub Repository: https://github.com/24eg105f30-lgtm/inksentinel-brand-governance • Solo Lead Engineer: Team Sentinel (24eg105f30@anurag.edu.in).